

The Impact of U.S. Tariff Changes on the German Economy.

Report: Data Analysis Project

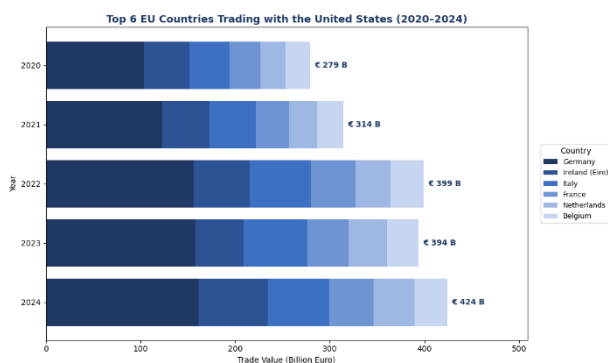
Authors: [Ifta Kakazai](#), [Hamza Boukhatem](#), [Mame Ndiaye](#) | [GitHub-Repository](#)

Abstract

This project examines the trade and macroeconomic consequences of the tariff measures introduced by the United States following the change of administration in 2024, with particular attention to their effect on Germany, the European Union's largest exporter to the U.S. market. Using trade, tariff, and macroeconomic datasets covering 2020–2026, we quantify the effective tariff burden across the United States' top trading partners, assess sectoral export exposure for Germany, and evaluate the broader impact on economic uncertainty, GDP growth, and the labor market.

1. Introduction

In 2025, the United States implemented a series of tariff increases affecting most of its major trading partners, including the European Union. These measures were intended to protect domestic industry, but they also raised import costs and introduced uncertainty for exporting nations. This project investigates how these tariffs affected European trade, using Germany as a case study. Germany was selected because it is consistently the EU's largest single exporter to the United States, trading more than double the value of the next-largest EU economy in every year between 2020 and 2024. Given this scale of exposure, Germany was expected to be the EU country most affected by the new tariff measures, making it the most informative case for assessing their real economic impact. The goal is to move beyond the political narrative surrounding tariffs and quantify their actual footprint in trade flows, sectoral exposure, and macroeconomic indicators.



2. Data and Methodology

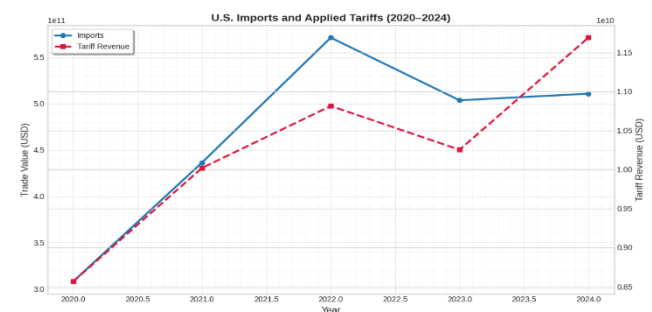
The analysis combines several datasets covering the period 2020–2026:

- Bilateral import and export values between the United States and its trading partners at the product level, sourced from the WTO/ADB trade indicators database;
- Tariff schedules under the Most Favored Nation (MTN) classification, distinguishing agricultural and non-agricultural products;
- Economic Policy Uncertainty (EPU) indices for Germany and the United States;
- Quarterly real GDP for Germany; and
- Employment growth by industry for Germany.

Import and tariff data were merged on year and product code to construct an implied **tariff value** (import value × tariff rate) for each partner, from which an **effective tariff rate** was derived. Trade and macroeconomic series were split into a pre-tariff period (2020–2024) and a post-tariff period (2025–2026) to compare distributions before and after the policy change.

3. U.S. Import Structure and Tariff Burden

Between 2020 and 2024, total U.S. imports across the analyzed products rose from approximately \$308 billion to \$511 billion, while the implied tariff revenue collected on these flows grew from \$8.6 billion to \$11.7 billion, an increase of roughly 36%. Canada, Mexico, and the European Union remained the largest suppliers to the U.S. market, together accounting for over half of the import value considered.

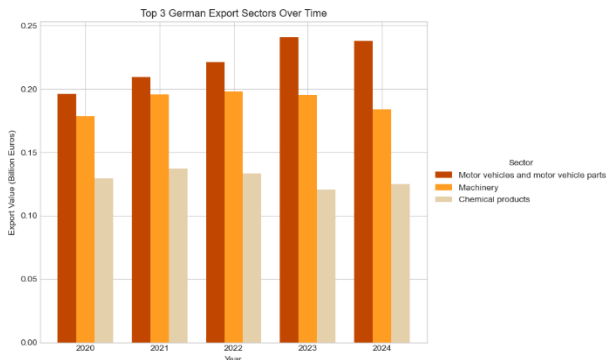


Interestingly, the size of a trading relationship does not translate directly into tariff exposure. Although Canada and Mexico account for the largest import shares, their effective tariff rates (1.40% and 2.66%) are markedly lower than that faced by the European Union (3.10%) and Brazil (3.40%), which is likely to reflect the preferential treatment historically extended to Canada and Mexico under regional trade arrangements. This indicates that the European Union, despite being a smaller supplier by volume, absorbed a proportionally heavier tariff burden.

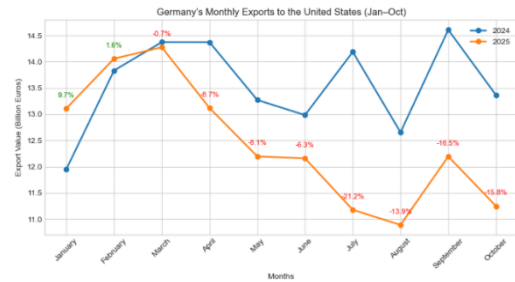
Trading Partner	Share of U.S. Imports	Import Value (2020–24)	Effective Tariff Rate
Canada	34.2%	\$796.7 B	1.40%
Mexico	13.3%	\$310.0 B	2.66%
European Union	10.2%	\$237.7 B	3.10%
Brazil	2.9%	\$66.7 B	3.40%
Saudi Arabia	2.8%	\$65.0 B	0.56%

4. Sectoral Trade Exposure: The Case of Germany

Motor vehicles and motor vehicle parts consistently represent Germany's largest export category to the United States, ahead of machinery and chemical products throughout 2020–2024. Exports in this sector grew steadily until 2023 but declined in 2024, and 2025 data confirm that motor vehicles remain the leading export category, followed by pharmaceuticals, automotive parts, and machinery. This concentration means that Germany's overall exposure to U.S. tariff policy is closely tied to the fortunes of its automotive sector.

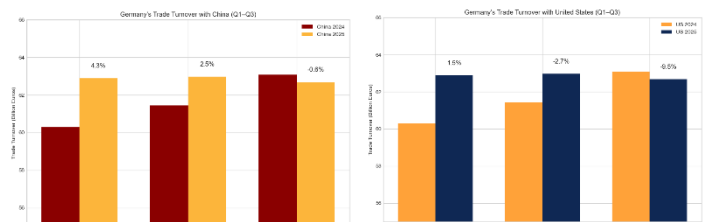


Monthly trade data for 2024–2025 show that German exports to the U.S. initially grew (+9.7% in January and +1.6% in February 2025, year-on-year) before tariff measures took effect. Once tariffs were announced, exports declined sharply from April onward, reaching a low of –21.2% in July 2025. Losses moderated later in the year as EU–U.S. negotiations settled on a reduced baseline tariff of around 15%, suggesting a partial, but incomplete, recovery.



5. Trade Reallocation: China Overtakes the United States

A further consequence of the tariff dispute was a shift in Germany's trading relationships. In 2025, China overtook the United States as Germany's largest trading partner, measured by quarterly trade turnover (exports plus imports). German Chinese turnover rose by 4.3% in Q1 2025 relative to Q1 2024 and by 2.5% in Q2, before slightly declining (–0.6%) in Q3. Over the same quarters, turnover with the United States moved from +1.5% in Q1 to –2.7% in Q2 and –9.5% in Q3, confirming a clear and accelerating divergence between the two relationships.

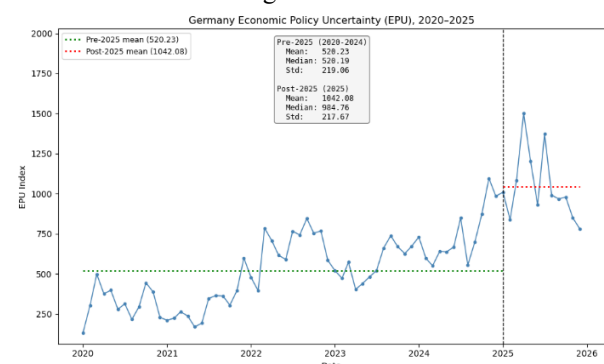


Trading Partner Trade Turnover 2025 (Billion €)

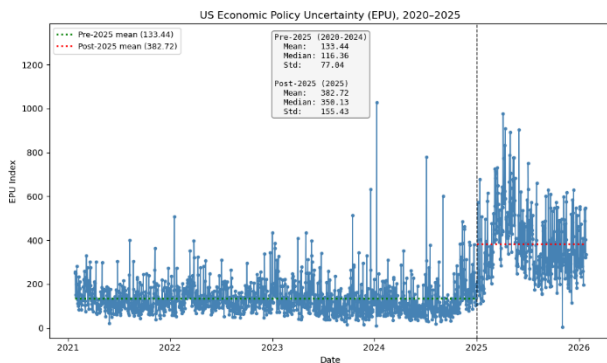
1	China	188.54
2	United States	185.23
3	Netherlands	158.34
4	France	138.35
5	Poland	135.30

6. Macroeconomic Effects

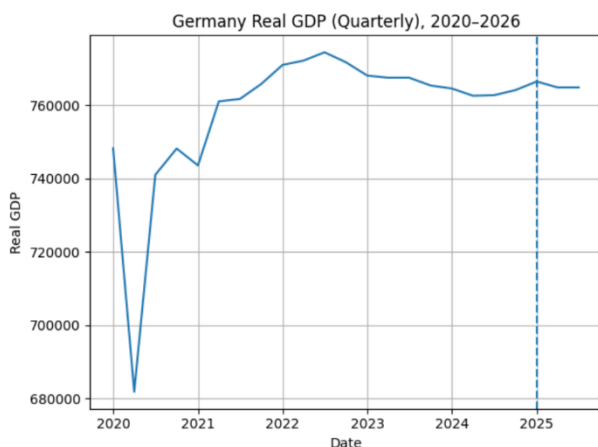
Beyond trade flows, the tariff episode coincided with a marked rise in economic policy uncertainty. The Economic Policy Uncertainty (EPU) Index measures how often media, forecasters, and policymakers' reference unpredictable government decisions; a higher index value signals greater uncertainty about future policy direction, which tends to discourage business investment and hiring.



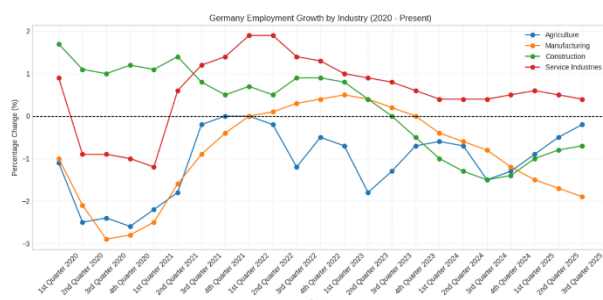
The U.S. EPU index nearly tripled over the period studied, from a mean of 133 in 2020–2024 to 383 in 2025–2026, reflecting the sharp rise in unpredictability introduced by the new tariff regime.



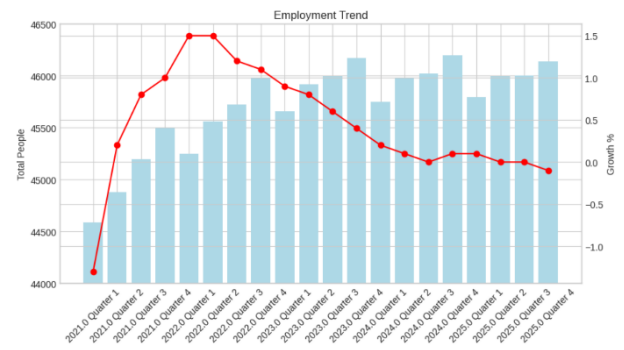
In Germany, quarterly real GDP growth slowed from an average of 0.15% before 2025 to 0.03% afterward, with volatility also falling sharply (standard deviation dropping from about 3.0 to 0.3). This points to a plateau in economic activity rather than an outright contraction, consistent with a resilient but weakened growth path.



Employment patterns show a similar shift in momentum. Before the tariffs, most sectors recovered steadily from the 2020 downturn: construction and services led the rebound, while manufacturing and agriculture turned positive by early 2022, and overall employment growth peaked at around 1.5% year-on-year in late 2021. After tariffs took effect in 2025, this momentum faded: manufacturing employment growth turned sharply negative again, falling to around –1.9% by the third quarter of 2025, and construction followed a similar downward path, while service industries proved comparatively more resilient, staying mildly positive. In aggregate, total employment kept rising slightly in



absolute terms, but year-on-year growth flattened to close to 0% through 2024–2025, pointing to a labour market that held onto its job base but lost the pace of expansion it had before the tariffs.



7. Conclusion

The findings show that the 2025 U.S. tariff measures had a measurable, though contained, impact on European trade. The tariff burden fell disproportionately on the European Union relative to its trading volume, Germany's export exposure remained concentrated in the automotive sector, and monthly trade data reveal a sharp contraction in mid-2025 followed by partial stabilization once a negotiated tariff ceiling was reached. In parallel, Germany's trade relationships shifted meaningfully toward China, while heightened policy uncertainty in the U.S. and a slowdown in German GDP growth confirm that the effects extended beyond bilateral trade figures into broader macroeconomic performance.

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